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SUNNY PANCHANI

R&D / Project Engineer — Machine Design & Automation

Ahmedabad, Gujarat, India • +91-8866605381 • sunnypanchani9007@gmail.com
https://www.linkedin.com/in/sunnypanchani-72b295117 • Portfolio : https://sunnypanchani.github.io

PROFESSIONAL SUMMARY

Project/R&D Engineer (Mechanical) with 10+ years of end-to-end project ownership across custom machine design, SPM development, industrial automation support, and on-site commissioning. Proven track record leading projects from client brief through concept design, structural and mechanical calculations, fabrication, documentation, testing, and final customer acceptance — across induction heating systems, grinding machines, and special-purpose equipment. Proficient in SolidWorks, AutoCAD, and Autodesk Inventor with strong capability in generating complete technical documentation packages including drawings, BOMs, inspection records, and O&M manuals. Experienced in supervising and coordinating cross-functional teams across design, production, and site; managing schedules, costs, and client expectations throughout the full project lifecycle. Hands-on expertise in retrofit engineering, hydraulic and pneumatic system integration, structural calculations for fabricated structures, and design-for-manufacture — consistently delivering 55–70% cost reductions versus imported machines while maintaining industrial-grade precision. Currently expanding technical capabilities into Python-based engineering automation and ANSYS-based simulation (FEA) to complement hands-on machine design and structural calculation expertise.

CORE COMPETENCIES

- End-to-End Project Execution
- Retrofit & Automation Engineering
- BOM & Drawing Package Generation
- Installation & Commissioning
- Production-to-Installation Coordination
- Hydraulic & Pneumatic Integration
- Structural & Mechanical Calculations
- Machine Design & SPM Development
- Client Liaison & Acceptance
- Manufacturing & Testing Coordination
- Cost & Schedule Management
- Technical Documentation
- Industrial Automation Support
- Engineering Automation & CAD Automation

TECHNICAL SKILLS

CAD & Design Software

Software: SolidWorks (10+ years), AutoCAD (10+ years), Autodesk Inventor (1+ year)

Design: 3D Modelling & Mechanism Design, Sheet Metal & Fabrication Design, Machine Design & Product Development, Industrial Automation & Retrofit Design, Hydraulic & Pneumatic System Integration, Manufacturing Drawings & Assembly Design, Complete Project Design & Production Coordination

Structural Calculation: Basic mechanical and structural calculations for machine components, weldments, and fabricated structures

Programming & Simulation (Learning)

Python (Mechanical Engineering Applications – Learning): Building working proficiency in Python for engineering calculations, CAD/BOM automation, and basic data analysis using Pandas and NumPy

ANSYS Mechanical (Simulation & FEA – Learning): Developing foundational skills in Static Structural Analysis, FEA fundamentals, stress and deformation evaluation, and introductory modal analysis

Engineering Automation: Applying scripting fundamentals to streamline repetitive design, BOM, and documentation workflows

CAD Automation: Exploring automation of part-numbering, BOM generation, and drawing data workflows within CAD software

Project & Documentation

Skills: Project Lifecycle Management, Post-Award Technical Documentation, BOM Generation & Control, Drawing Package Preparation, Inspection & Test Plans (ITP), Manufacturing & As-Built Documentation, O&M Manual Preparation, Material Data Review & Control, MS Office (Excel, Word, PowerPoint), Project Scheduling & Cost Tracking

Manufacturing & Field

Skills: CNC/VMC Manufacturing Coordination, Machining Process Understanding, Production & Vendor Coordination, Welding, Fabrication & Assembly, Machine Alignment & Setup, Hydraulics & Pneumatics, Test Procedure Development, Product Validation & Quality Control, Root Cause Analysis, Breakdown Troubleshooting

Standards & Process

Skills: ISO Quality Systems (exposure), IEC Standards — Industrial Equipment, SPM Design & Development, Retrofit & Automation Engineering, Design for Manufacture (DFM), Lean Manufacturing & Waste Reduction, Six Sigma Exposure, Cost Optimisation & Value Engineering

PROFESSIONAL EXPERIENCE

R&D Engineer (R&D & Project Engineering) | Armstrong Machinery LLP

June 2026 – Present • Vatva GIDC, Ahmedabad

- Handling combined R&D and Project Engineering responsibilities, covering new machine development alongside ongoing production-line engineering support.
- Design and development of new machinery and automation systems.
- 3D modeling, assemblies, manufacturing drawings, and BOM creation using Autodesk Inventor and SolidWorks.
- Machine frame, weldment, and sheet metal design.
- Basic mechanical and structural calculations for machine components and fabricated structures.
- Design for Manufacturing (DFM) and material selection.
- Developing a standardized part-numbering and BOM automation system for the production line — structured part codes (e.g., AR051-110-01-C format) with descriptions auto-populated from CAD software, replacing the current manual BOM, material, and weight documentation process.
- Creating Standard Operating Procedures (SOPs) for junior design engineers to standardize workflows in the design department.
- Coordination with fabrication, welding, machining, production, and assembly teams.
- Prototype development, testing, troubleshooting, and design improvements.
- Engineering documentation, drawing revisions, and BOM management.
- Support installation, commissioning, and continuous product improvement.

Project Engineer (Mechanical) | Interpower Induction Pvt. Ltd.

Dec 2024 – June 2026 • Santej, Ahmedabad | Induction Heating Systems — Automotive, Bearing & Fastener Industries

- Led and managed complete project lifecycle of custom induction heating machine systems — from initial client requirement discussions and concept design through fabrication, pre-dispatch verification, on-site installation, commissioning, and formal customer acceptance.
- Produced full mechanical design and drawing packages in Autodesk Inventor and AutoCAD — including Exit Conveyor, Loading Automation, Coil Shuttle, Heat Station, Power Supply Enclosure, and DM Unit — with all production-ready documentation issued and controlled throughout each project.
- Generated and managed post-award technical documentation including detailed drawings, BOMs, inspection records, and layout updates; maintained accurate as-built records and reissued revised drawing sets to production when engineering changes were required.
- Supervised and coordinated cross-functional teams across design, manufacturing, and site execution throughout each project; managed schedules, resolved technical issues on the floor, and ensured design intent was maintained during fabrication.
- Conducted pre-dispatch quality and layout verification — reviewed components against drawings and specifications before shipment, identified non-conformances, and drove resolution prior to dispatch.
- Executed on-site machine installation and commissioning — mechanical alignment, system checks, parameter setting, and customer acceptance trials — delivering machines fully operational at handover.
- Managed direct client communication throughout the project lifecycle; provided post-commissioning support, operator training, and on-site engineering modifications as required by the customer.
- Delivered end-to-end projects for automotive, bearing, and fastener industry customers with full ownership from first drawing to commissioned handover.

Project Engineer | Machine Design & Manufacturing | SR Tool Line

Nov 2019 – Nov 2024 • Kathwada GIDC, Ahmedabad

- Led end-to-end development of new SPM and custom machine projects — managing concept design, mechanism development, fabrication coordination, assembly, functional testing, and commissioned delivery to customers.
- Designed and delivered TCT Saw Grinding Machines (Side, Top, and Face Grinding variants) achieving 5–7 micron precision; led complete project from SolidWorks design through calibration, test validation, and formal customer handover.
- Managed full project lifecycle of Glass Edge Chamfering Machine — design, fabrication, assembly, alignment, and commissioned prototype delivered within a 30-day hard deadline for industrial exhibition.
- Designed and executed Automatic Hydraulic Top Grinding Machine — full mechanism and hydraulic circuit design, structural analysis, BOM generation, and complete manufacturing documentation package; achieved 55% cost reduction versus imported equivalent machines.
- Developed and executed test procedures for all new machine designs; supervised product validation and assembly compliance checks against specification prior to delivery; compiled handover documentation packages for each project.
- Prepared and controlled full manufacturing documentation sets including drawings, BOMs, operation manuals, and maintenance guides; managed drawing revisions throughout project execution to ensure production teams always worked to current specifications.
- Coordinated cross-functional teams of vendors, CNC/VMC operators, and fabricators throughout each project; supervised material procurement, production scheduling, quality verification, and on-time delivery against project plan.
- Delivered all machines with 55–70% cost reduction versus imported equivalents while maintaining industrial-grade quality and precision — demonstrating strong design-for-manufacture and value engineering capability.

Manufacturing Engineer | New Machine Development | Yash Tooling System
Dec 2015 – Oct 2019 • Kathwada GIDC, Ahmedabad

- Led end-to-end new machine development projects for custom woodworking tool manufacturing machinery — managing requirement analysis, concept design, fabrication coordination, assembly, and commissioning.
- Designed custom SPM grinding, cutting, and sharpening machines for carbide-tipped saw blade production using AutoCAD; delivered complete controlled project packages including drawings, BOMs, and operation manuals.
- Supervised and coordinated CNC/VMC operators, fabricators, and fitters across concurrent project stages to ensure design intent, dimensional quality, and on-time delivery.
- Managed project cost estimation, material procurement, vendor coordination, and production scheduling across multiple simultaneous machine development projects.

Production Engineer | Hi-Tech Industry
Jul 2014 – Dec 2015 • Kathwada GIDC, Ahmedabad

- Supervised daily production operations; coordinated output targets, quality standards, and delivery schedules across the production floor.
- Led design modification and equipment improvement projects using AutoCAD; provided manufacturing feasibility inputs and cost estimates for new product development initiatives.
- Maintained controlled production documentation, quality records, and equipment maintenance schedules.

KEY PROJECTS

Induction Heating Machine Systems — Interpower (2024–2026): Led full project ownership — client brief, concept design, drawing packages, BOM, pre-dispatch checks, on-site installation, commissioning, and customer handover for automotive and bearing industry clients. Supervised cross-functional coordination across design, fabrication, and site teams.

Automatic Hydraulic Top Grinding Machine — SR Tool Line (2024): Managed end-to-end SPM project — mechanism & hydraulic circuit design, full documentation package, fabrication coordination, test procedure development, calibration, and commissioned delivery. Achieved 55% cost saving vs. imported machines with 5–7 micron precision.

Glass Edge Chamfering Machine — SR Tool Line (Exhibition Model): Led complete project delivery within a 30-day hard deadline — design, fabrication, assembly, alignment, test validation, and commissioned prototype for live industrial exhibition. On-time delivery with full quality compliance.

TCT Saw Grinding Machine (Side, Top & Face Variants) — SR Tool Line: Designed and delivered complete SPM product family from SolidWorks models to commissioned handover. Achieved 70% cost reduction vs. market machines with 5–7 micron accuracy. Designed for low-skill operator use with complete O&M documentation package.

Woodworking Tool SPM Series — Yash Tooling System (2015–2019): Led development of multiple custom SPMs for carbide-tipped saw blade manufacturing — grinding, cutting, and sharpening machines designed from scratch in AutoCAD with full drawing packages, BOMs, and operation manuals. Supervised fabrication, assembly, and commissioning across all projects.

LEARNING & TECHNICAL DEVELOPMENT

- Currently developing working proficiency in Python for mechanical engineering applications, with focus on automation of engineering calculations, CAD/BOM data handling, and basic data analysis using Pandas and NumPy.
- Building foundational skills in ANSYS Mechanical for Finite Element Analysis (FEA), including Static Structural Analysis, stress and deformation evaluation, and introductory modal analysis.
- Developing working knowledge of engineering automation techniques to streamline documentation, BOM generation, and repetitive design workflows — applied practically in current part-numbering and BOM automation initiative.
- Growing interest in simulation-driven design and design validation methods to complement hands-on manufacturing and structural calculation experience.
- Pursuing self-directed, structured learning through online courses and practical exercises to build a foundation for applying computational and simulation tools alongside traditional CAD-based machine design.

EDUCATION

Bachelor of Engineering (B.E.) — Mechanical Engineering | 2014
Kalol Institute of Technology and Research Center, Gujarat, India • CGPA: 6.67 / 10.0

CERTIFICATIONS & ACHIEVEMENTS

- Lean Manufacturing & Six Sigma Exposure — Process Improvement & Quality Management
- Machine Design Standards & Manufacturing Practices — ISO Quality Systems & IEC Industrial Equipment
- Led 15+ custom SPM and machine projects from concept to commissioned delivery across automotive, bearing, fastener, and woodworking tool industries
- Achieved 55–70% cost reduction versus imported machines while maintaining industrial-grade precision across all delivered projects
- Delivered exhibition-critical machine project within a 30-day hard deadline — design through commissioned prototype

LANGUAGES

English: Professional Working Proficiency • Hindi: Native / Bilingual • Gujarati: Native / Bilingual